MBA PROJECT BRIEF — UFT ROTMAN

Reclaimed Lumber intelligence for Circular Construction Canada

Developed by professional MBAs at the University of Toronto Rotman School of Management

How can Circular Construction Canada transform fragmented demolition and salvage data on wood from the construction, renovation, and demolition (CRD) sector into dynamic, ecosystem-wide intelligence that enables stakeholders to forecast potentially recoverable reclaimed lumber supply, identify recovery bottlenecks, and coordinate circular material flows across Canada's 25 largest metropolitan regions?

THE ORGANIZATION

Circular Construction Canada, a Solution Space of the non-profit charity Generate Canada, is working to accelerate the adoption of circular strategies and practices across Canada's construction and real estate sector.

Through its Circular Construction Innovation Hub, the organization convenes industry and policy leaders, supports regional demonstration projects and applied research, and shares practical knowledge to advance affordable, market-ready solutions. Its work helps de-risk investment in circular strategies, practices, and technologies while building the conditions for wider adoption across the sector.

Wood Reuse in Construction Project

Circular Construction Canada's Wood Reuse in Construction project, supported by the Max Bell Foundation, is developing a practical industry guide and policy briefs to help advance the safe, scalable use of salvaged and reclaimed wood in Canada's construction sector. The project combines research, testing coordination, demonstration projects, and stakeholder engagement to reduce technical, policy, and market barriers while supporting future standards, grading protocols, and circular construction practices.

THE CHALLENGE

Reclaimed wood has significant reuse potential, but several barriers prevent it from becoming a reliable construction material at scale, including:

- **Lack of standardized grading methods:** There is no clear, scalable pathway for confirming the quality, durability, and performance of reclaimed wood, especially for structural reuse.
- **High recovery and one-off certification costs:** Deconstruction, sorting, denailing, testing, and project-by-project engineering approvals can make reclaimed wood harder and more expensive to use than virgin lumber.
- **Fragmented supply chains:** Demolition and deconstruction contractors, processors, designers, builders, and reuse buyers are not well connected, making it difficult to move salvageable wood from demolition sites into new projects.
- **Insufficient recovery requirements and infrastructure:** Areas with high demolition activity often lack the rules, processing capacity, storage, and logistics needed to capture salvageable material before it is landfilled.
- **No reliable forward supply signal:** Processors and reuse buyers often do not know where, when, or in what condition salvageable wood will become available, making it difficult to plan inventory or projects.
- **Persistent market mismatch:** Usable lumber goes to landfill not because it lacks value, but because the system lacks coordination between demolition supply and construction demand.
- **Uncertainty around durability, quality, cost, and availability:** Without reliable information and standards, project teams view reclaimed wood as risky, discouraging reuse at scale.

Residential demolition is a major and recurring source of salvageable lumber, especially where older stick-frame housing is being replaced through infill redevelopment. Prior research in the City of Toronto found that a large share of demolition permits were for single-family homes, with post-war houses from the 1950s and 1960s appearing as a common demolition typology. These buildings are typically stick-frame construction and can contain significant quantities of reusable dimensional lumber. That being said, larger, older and more sought after wood types for reuse, such as old-growth Douglas fir beams versus 2 X 4s, have more value on the existing reuse market due the reuse challenges explained above.

The opportunity is also localized because demolition activity tends to cluster in specific neighbourhoods, redevelopment corridors, and housing typologies rather than being evenly distributed across a city. This means the intelligence layer with higher confidence should identify neighbourhood-level demolition hotspots, and common building archetypes.

Policy Ambition Without Operational Capacity

Many municipalities have adopted circular economy or zero-waste targets, yet lack the data infrastructure to operationalize them. Without a clear picture of material flows, recovery rates, and ecosystem gaps, policy commitments remain aspirational rather than actionable.

The core challenge is to replace fragmented, point-in-time estimates with a dynamic, transparent, and defensible intelligence layer that continuously maps where salvageable lumber could emerge, what recovery capacity exists, and where the critical bottlenecks and opportunities lie.

WHAT WE WILL BUILD

We will build a salvageable lumber intelligence layer for Canada's 25 largest metropolitan regions — combining open-source data pipelines, ecosystem mapping, and decision-support tooling to provide Circular Construction Canada with a proactive ecosystem awareness, data-driven solution.

- **Municipal demolition and housing stock baseline:** A city-wide model profiling demolition permit trends, housing stock age, building typology, and regional wood construction patterns across the 25 target metropolitan regions (CMAs). This establishes a first-order estimate of how much salvageable wood may exist in each market.
- **Neighbourhood hotspot and archetype refinement:** A targeted geospatial analysis within priority municipalities that identifies where demolition activity is concentrated and which repeatable building types are being removed. This refines the model from a broad city-wide estimate into a more realistic recoverable supply estimate by adjusting for local demolition clusters, housing age, building form, wood-frame likelihood, and practical recovery conditions.
- **Material flow and recovery rate estimates:** Confidence-scored forecasts of potential salvageable lumber volumes by municipality, neighbourhood, building type, age cohort, and material category. The model should attempt to distinguish between gross wood content, recoverable wood, salvageable dimensional lumber, non-structural material, finish material, and specification-ready reusable lumber. It should be able to forecast over a 5–10 year horizon, incorporating demolition trends, recovery assumptions, and degradation factors.
- **Ecosystem actor map and gap analysis:** A structured mapping of the full circular lumber supply chain —demolition contractors, deconstruction specialists, salvage processors, warehouses, and reuse buyers — against the supply forecast. Key outputs include identification of municipalities with strong demolition activity but weak recovery infrastructure, material supply nodes with no visible downstream buyers, and geographic whitespace in processing and warehousing capacity.
- **Coordination platform architecture and roadmap:** A strategic blueprint for evolving the intelligence layer into a live coordination and decision-support platform — enabling

real-time permit monitoring, matchmaking between supply and demand nodes, and dynamic recovery forecasting. Includes a build versus partner analysis and integration pathway with existing municipal and industry data systems.

Nice to Have Criteria

- **Policy and incentive alignment matrix:** An analysis mapping municipal circular economy commitments and procurement policies against operational ecosystem capacity, identifying where policy ambition outpaces infrastructure and where targeted interventions (procurement mandates, salvage requirements in demolition permits, tax incentives) would generate the greatest material recovery uplift.
- **Storage and logistics capacity layer:** A practical map of the infrastructure needed to make reuse possible, including covered yards, reuse hubs, warehouse space, industrial lands, municipal assets, port lands, processors, and temporary staging areas. This is critical because storage is often the missing link between salvage and reuse.
- **Portfolio-scale reuse opportunity scan:** Identification of public agencies, large landowners, campuses, ports, airports, housing authorities, and redevelopment agencies that control both demolition and construction activity. These actors may be best positioned to close the loop internally by matching materials from one site to another within the same portfolio.
- **Foundry model and long-term asset strategy:** A strategic framework for extending the ecosystem intelligence methodology beyond salvaged lumber to other materials and/or activities (e.g., adaptive reuse or home relocation) — establishing the proprietary data and network asset that creates a durable competitive advantage for CCC as a national circular economy infrastructure platform.
- **Circular Construction Resource and Matchmaking Hub:** We are exploring whether Circular Construction Canada could support an online hub for circular construction resources and possibly a stakeholder matchmaking platform for Canada’s circular construction and real estate sector.